THE INJECTIVE HULL OF THE RESIDUE FIELD BASE CHANGES ALONG A FLAT LOCAL HOMOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the hull as the union of the submodules killed by the powers of the maximal ideal. Invoked at the start of the counted proof (source line 5521). Reduces the statement to the truncated pieces over the Artinian quotients.
- The equality of the lengths of the hull and of the ring over an Artinian local ring. Invoked inside the counted proof (source lines 5528-5531). Compares the length of the base changed hull with that of the hull of the target residue field.
- The criterion for an injective module map between hulls to be an isomorphism. Invoked inside the counted proof (source lines 5540-5542). Produces the map and upgrades it to an isomorphism once the lengths agree.
- The equality of lengths under the flat pullback of finite length modules. Invoked inside the counted proof (source lines 5525-5527). Makes the length of the base changed module computable over the target.

Source reference. Original reference algebra-lemma-pullback-module: sources/stacks-dualizing/pinned-original-source.tex, source lines 5527–5527. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-pullback-module, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-finite: sources/stacks-dualizing/pinned-original-source.tex, source lines 912–912. Navigation pointer only: the counted text refers to the Stacks item lemma-finite, which is an existing labeled result of this

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same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-torsion-submodule-sum-injective-hulls: sources/stacks-dualizing/pinned-original-source.tex, source lines 1023–1023. Navigation pointer only: the counted text refers to the Stacks item lemma-torsion-submodule-sum-injective-hulls, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-union-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1042–1042. Navigation pointer only: the counted text refers to the Stacks item lemma-union-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa) \to (B, \mathfrak{n}, l)$ be a flat local homorphism of Noetherian rings such that $\mathfrak{n} = \mathfrak{m}B$. If E is the injective hull of κ, then $E \otimes_A B$ is the injective hull of l.*

Proof. Write $E = \bigcup E_n$ as in Lemma lemma-union-artinian. It suffices to show that $E_n \otimes_{A/\mathfrak{m}^n} B/\mathfrak{n}^n$ is the injective hull of l over $B/\mathfrak{n}$. This reduces us to the case where A and B are Artinian local. Observe that $\text{length}_A(A) = \text{length}_B(B)$ and $\text{length}_A(E) = \text{length}_B(E \otimes_A B)$ by Algebra, Lemma algebra-lemma-pullback-module. By Lemma lemma-finite we have $\text{length}_A(E) = \text{length}_A(A)$ and $\text{length}_B(E') = \text{length}_B(B)$ where E' is the injective hull of l over B. We conclude $\text{length}_B(E') = \text{length}_B(E \otimes_A B)$. Observe that

$$\dim_l((E \otimes_A B)[\mathfrak{n}]) = \dim_l(E[\mathfrak{m}] \otimes_A B) = \dim_\kappa(E[\mathfrak{m}]) = 1$$

where we have used flatness of $A \to B$ and $\mathfrak{n} = \mathfrak{m}B$. Thus there is an injective B-module map $E \otimes_A B \to E'$ by Lemma lemma-torsion-submodule-sum-injective-hulls. By equality of lengths shown above this is an isomorphism. □